REDUCEDNESS ASCENDS ALONG A REGULAR MAP FROM A REDUCED NOETHERIAN RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A Noetherian ring is reduced if and only if it satisfies (S_1) and (R_0). Algebra, Lemma \ref{algebra-lemma-criterion-reduced}, cited at proof lines 10544-10546. Replaces reducedness by the two local conditions.
- The conditions (S_k) and (R_k) ascend along regular ring maps. Algebra, Lemma \ref{algebra-lemma-Sk-goes-up} and Algebra, Lemma \ref{algebra-lemma-Rk-goes-up}, cited at proof lines 10547-10549. Give the two conditions for the target.

Source reference. Original lemma 07QK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9835–9835. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 031R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3994–3994. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0339: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4048–4048. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4049–4049. Navigation pointer to a statement the

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Lemma 0.2. *Let $\varphi : R \to S$ be a ring map. Assume*

(1) *φ is regular,*
(2) *S is Noetherian, and*
(3) *R is Noetherian and reduced.*

Then S is reduced.

Proof. For Noetherian rings being reduced is the same as having properties (S_1) and (R_0), see Algebra, Lemma 031R. Hence we may apply Algebra, Lemmas 0339 and 033A. □